MPG No. 9.3.31 $4'$ [Type III, tetragonal] [M tensor]

* Rank 0 tensor. * Rank 1 tensor. * Rank 2 tensor (s).

$$\begin{bmatrix} M_{xx} & M_{xy} & 0 \\ M_{xy} & -M_{xx} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{xx} = \mathbb{M}_{v}^{(1)}$$
$$M_{xy} = \mathbb{M}_{xy}^{(1)}$$

* Rank 2 tensor (a). * Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & 0 & M_{xxz} \\ 0 & 0 & -M_{xxz} \\ 0 & 0 & 0 \\ M_{yzx} & -M_{xxz} & 0 \\ M_{xxz} & M_{yzx} & 0 \\ 0 & 0 & M_{yzx} \end{bmatrix}$$

$$M_{xxz} = \mathbb{M}_{z}^{\beta(1)}$$
$$M_{yzx} = \mathbb{M}_{xyz}^{(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & M_{xxz} \\ 0 & 0 & -M_{xxz} \\ 0 & 0 & 0 \\ M_{yzx} & \frac{M_{xxz}}{2} & 0 \\ -\frac{M_{xxz}}{2} & M_{yzx} & 0 \\ 0 & 0 & -2M_{yzx} \end{bmatrix}$$

$$M_{xxz} = 2\mathbb{T}_{xy}^{(1)}$$
$$M_{yzx} = \mathbb{T}_{v}^{(1)}$$
$$\frac{M_{xxz}}{2} = \mathbb{T}_{xy}^{(1)}$$
$$2M_{yzx} = 2\mathbb{T}_{v}^{(1)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} M_{yzx} & M_{yzy} & 0 \\ M_{yzy} & -M_{yzx} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzx} = \mathbb{T}_{v}^{(2)}$$
$$M_{yzy} = \mathbb{T}_{xy}^{(2)}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} M_{xxxx} & 0 & M_{xxzz} & 0 & 0 & M_{xxxy} \\ 0 & -M_{xxxx} & -M_{xxzz} & 0 & 0 & M_{xxxy} \\ M_{xxzz} & -M_{xxzz} & 0 & 0 & 0 & M_{zzxy} \\ 0 & 0 & 0 & M_{yzyz} & M_{yzzx} & 0 \\ 0 & 0 & 0 & M_{yzzx} & -M_{yzyz} & 0 \\ M_{xxxy} & M_{xxxy} & M_{zzxy} & 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxxx} = -\mathbb{M}_{4v}^{(1)} + 2\mathbb{M}_{v}^{(1)} + 4\mathbb{M}_{v}^{(2)}$$
$$M_{xxzz} = \mathbb{M}_{4v}^{(1)} + \mathbb{M}_{v}^{(1)}$$
$$M_{xxxy} = -\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)}$$
$$M_{zzxy} = 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)}$$
$$M_{yzyz} = -\mathbb{M}_{4v}^{(1)} - \mathbb{M}_{v}^{(2)}$$
$$M_{yzzx} = 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(2)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & M_{xxyy} & M_{xxzz} & 0 & 0 & M_{xxxy} \\ -M_{xxyy} & 0 & -M_{xxzz} & 0 & 0 & M_{xxxy} \\ -M_{xxzz} & M_{xxzz} & 0 & 0 & 0 & M_{zzxy} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -M_{xxxy} & -M_{xxxy} & -M_{zzxy} & 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxyy} = -2\mathbb{M}_{v}^{(3)} + 4\mathbb{T}_{xyz}^{(1)}$$
$$M_{xxzz} = -\mathbb{M}_{v}^{(3)} - 4\mathbb{T}_{xyz}^{(1)}$$
$$M_{xxxy} = \mathbb{M}_{xy}^{(3)} + 2\mathbb{T}_{z}^{\beta(1)}$$
$$M_{zzxy} = \mathbb{M}_{xy}^{(3)} - 4\mathbb{T}_{z}^{\beta(1)}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} M_{yzyz} & M_{yzzx} & 0 \\ M_{yzzx} & -M_{yzyz} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzyz} = 2\mathbb{M}_{v}^{(4)}$$
$$M_{yzzx} = 2\mathbb{M}_{xy}^{(4)}$$

* Rank 4 tensor (aaa). * Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & 0 & M_{xxxy} \\ 0 & 0 & -M_{xxxy} \\ 0 & 0 & 0 \\ M_{yzyz} & M_{yzzx} & 0 \\ -M_{yzzx} & M_{yzyz} & 0 \\ 0 & 0 & M_{xyxy} \end{bmatrix}$$

$$M_{xxxy} = -2\mathbb{M}_{xy}^{(5)} + \mathbb{T}_{z}^{\beta(2)}$$
$$M_{yzyz} = -\mathbb{M}_{v}^{(5)} + \mathbb{T}_{xyz}^{(2)}$$
$$M_{yzzx} = -\mathbb{M}_{xy}^{(5)} - \mathbb{T}_{z}^{\beta(2)}$$
$$M_{xyxy} = 2\mathbb{M}_{v}^{(5)} + \mathbb{T}_{xyz}^{(2)}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & M_{yzyz} & M_{yzzx} & 0 \\ 0 & 0 & 0 & -M_{yzzx} & M_{yzyz} & 0 \\ M_{xyxx} & -M_{xyxx} & 0 & 0 & 0 & M_{xyxy} \end{bmatrix}$$

$$M_{yzyz} = -\mathbb{M}_{v}^{(6)} + \mathbb{T}_{xyz}^{(3)}$$
$$M_{yzzx} = \mathbb{M}_{xy}^{(6)} + \mathbb{T}_{z}^{\beta(3)}$$
$$M_{xyxx} = -2\mathbb{M}_{xy}^{(6)} + \mathbb{T}_{z}^{\beta(3)}$$
$$M_{xyxy} = 2\mathbb{M}_{v}^{(6)} + \mathbb{T}_{xyz}^{(3)}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} M_{xxxx} & M_{xxyy} & M_{xxzz} & 0 & 0 & M_{xxxy} & 0 & 0 & M_{xxyx} \\ -M_{xxyy} & -M_{xxxx} & -M_{xxzz} & 0 & 0 & M_{xxyx} & 0 & 0 & M_{xxxy} \\ M_{zzxx} & -M_{zzxx} & 0 & 0 & 0 & M_{zzxy} & 0 & 0 & M_{zzxy} \\ 0 & 0 & 0 & M_{yzyz} & M_{yzzx} & 0 & M_{yzzy} & M_{yzxz} & 0 \\ 0 & 0 & 0 & M_{yzxz} & -M_{yzzy} & 0 & M_{yzzx} & -M_{yzyz} & 0 \\ M_{xyxx} & M_{xyxx} & M_{xyzz} & 0 & 0 & M_{xyxy} & 0 & 0 & -M_{xyxy} \end{bmatrix}$$

$$\begin{aligned}
M_{xxxx} &= -\mathbb{M}_{4v}^{(1)} + 2\mathbb{M}_{v}^{(1)} + 4\mathbb{M}_{v}^{(2)} \\
M_{xxyy} &= -2\mathbb{M}_{v}^{(3)} + 4\mathbb{T}_{xyz}^{(1)} \\
M_{xxzz} &= \mathbb{M}_{4v}^{(1)} + \mathbb{M}_{v}^{(1)} - \mathbb{M}_{v}^{(3)} - 4\mathbb{T}_{xyz}^{(1)} \\
M_{xxxy} &= -\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(3)} - 2\mathbb{M}_{xy}^{(5)} + 2\mathbb{T}_{z}^{\beta(1)} + \mathbb{T}_{z}^{\beta(2)} \\
M_{xxyx} &= -\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(3)} + 2\mathbb{M}_{xy}^{(5)} + 2\mathbb{T}_{z}^{\beta(1)} - \mathbb{T}_{z}^{\beta(2)} \\
M_{zzxx} &= \mathbb{M}_{4v}^{(1)} + \mathbb{M}_{v}^{(1)} + \mathbb{M}_{v}^{(3)} + 4\mathbb{T}_{xyz}^{(1)} \\
M_{zzxy} &= 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + \mathbb{M}_{xy}^{(3)} - 4\mathbb{T}_{z}^{\beta(1)} \\
M_{yzyz} &= -\mathbb{M}_{4v}^{(1)} - \mathbb{M}_{v}^{(2)} - \mathbb{M}_{v}^{(5)} + \mathbb{T}_{xyz}^{(2)} \\
M_{yzzx} &= 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(2)} - \mathbb{M}_{xy}^{(5)} - \mathbb{T}_{z}^{\beta(2)} \\
M_{yzzy} &= -\mathbb{M}_{4v}^{(1)} - \mathbb{M}_{v}^{(2)} + \mathbb{M}_{v}^{(5)} - \mathbb{T}_{xyz}^{(2)} \\
M_{yzxz} &= 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(5)} + \mathbb{T}_{z}^{\beta(2)} \\
M_{xyxx} &= -\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} - \mathbb{M}_{xy}^{(3)} - 2\mathbb{T}_{z}^{\beta(1)} \\
M_{xyzz} &= 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} - \mathbb{M}_{xy}^{(3)} + 4\mathbb{T}_{z}^{\beta(1)} \\
M_{xyxy} &= 2\mathbb{M}_{v}^{(5)} + \mathbb{T}_{xyz}^{(2)}
\end{aligned}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & M_{yzyz} & M_{yzzx} & 0 & M_{yzzy} & M_{yzxz} & 0 \\ 0 & 0 & 0 & -M_{yzxz} & M_{yzzy} & 0 & -M_{yzzx} & M_{yzyz} & 0 \\ M_{xyxx} & -M_{xyxx} & 0 & 0 & 0 & M_{xyxy} & 0 & 0 & M_{xyxy} \end{bmatrix}$$

$$\begin{aligned}
M_{yzyz} &= 2\mathbb{M}_{v}^{(4)} - \mathbb{M}_{v}^{(6)} + \mathbb{T}_{xyz}^{(3)} \\
M_{yzzx} &= 2\mathbb{M}_{xy}^{(4)} + \mathbb{M}_{xy}^{(6)} + \mathbb{T}_{z}^{\beta(3)} \\
M_{yzzy} &= -2\mathbb{M}_{v}^{(4)} - \mathbb{M}_{v}^{(6)} + \mathbb{T}_{xyz}^{(3)} \\
M_{yzxz} &= -2\mathbb{M}_{xy}^{(4)} + \mathbb{M}_{xy}^{(6)} + \mathbb{T}_{z}^{\beta(3)} \\
M_{xyxx} &= -2\mathbb{M}_{xy}^{(6)} + \mathbb{T}_{z}^{\beta(3)} \\
M_{xyxy} &= 2\mathbb{M}_{v}^{(6)} + \mathbb{T}_{xyz}^{(3)}
\end{aligned}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} M_{xxxx} & M_{xxxy} & 0 \\ M_{xxxy} & -M_{xxxx} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & M_{yyzz} \\ M_{zzxx} & M_{zzxy} & 0 \\ M_{xxyx} & -M_{yyzz} - M_{zzxx} & 0 \\ M_{yzzx} & M_{yzzy} & 0 \\ 0 & 0 & M_{zxxz} \\ -M_{yzzy} - M_{zxxz} & M_{xyyy} & 0 \\ 0 & 0 & -\frac{M_{xxxy}}{2} - \frac{M_{xxyx}}{2} + M_{xyyy} + M_{zzxy} \end{bmatrix}$$

$$M_{xxxx} = -\mathbb{M}_{4v}^{(1)} + 2\mathbb{M}_{v}^{(1)} + 4\mathbb{M}_{v}^{(2)}$$
$$M_{xxxy} = -\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(3)} - 2\mathbb{M}_{xy}^{(5)}$$
$$M_{yyzz} = -\mathbb{M}_{4v}^{(1)} - \mathbb{M}_{v}^{(1)} + \mathbb{M}_{v}^{(3)}$$
$$M_{zzxx} = \mathbb{M}_{4v}^{(1)} + \mathbb{M}_{v}^{(1)} + \mathbb{M}_{v}^{(3)}$$
$$M_{zzxy} = 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + \mathbb{M}_{xy}^{(3)}$$
$$M_{xxyx} = -\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(3)} + 2\mathbb{M}_{xy}^{(5)}$$
$$M_{yzzx} = 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(2)} - \mathbb{M}_{xy}^{(5)}$$
$$M_{yzzy} = -\mathbb{M}_{4v}^{(1)} - \mathbb{M}_{v}^{(2)} + \mathbb{M}_{v}^{(5)}$$
$$M_{zxxz} = \mathbb{M}_{4v}^{(1)} + \mathbb{M}_{v}^{(2)} + \mathbb{M}_{v}^{(5)}$$
$$M_{xyyy} = -\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} - \mathbb{M}_{xy}^{(3)}$$